



Fig. 6: Digital clock with temperature display

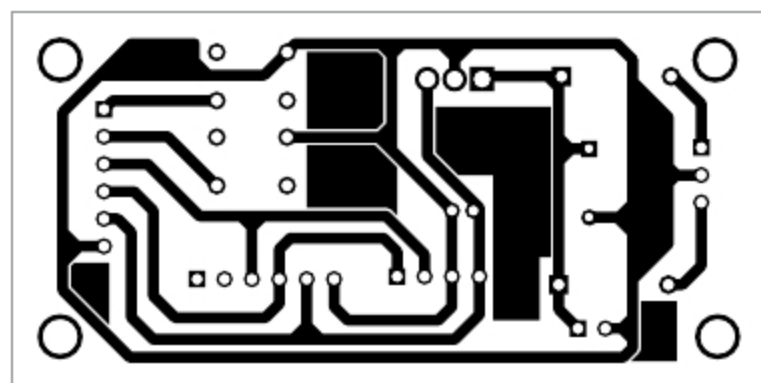


Fig. 7: Actual-size PCB layout for real-time clock with temperature display

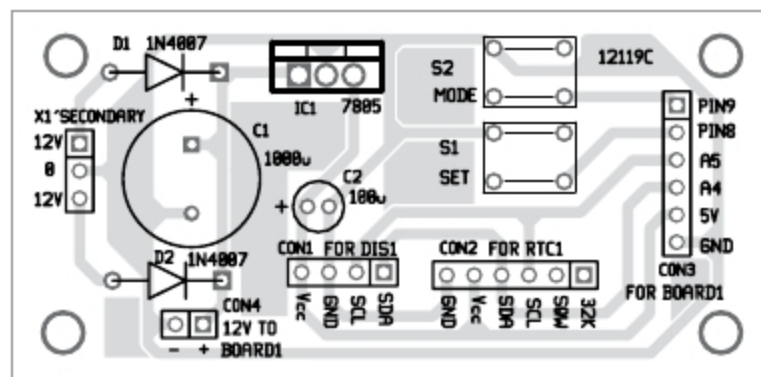


Fig. 8: Components layout of the PCB

minutes quickly, or do a short press to advance to the next setting. Press and release the S2 button to advance the cursor to the seconds' field. Press S2 button momentarily to reset the seconds to zero, or hold the button to freeze the seconds at zero and release to synchronise with an external time

source. DS3231 RTC chip has an inbuilt temperature sensor. Thus, it gives time, date, and year along with temperature readings (refer Fig. 6).

Software

Circuit operation is controlled by the software program (main_code.ino) loaded into the internal memory of Arduino Uno. The program implements all the required functionalities. The program/sketch is written in the Arduino programming language. Arduino IDE version 1.8.11 is used to compile and upload the program to the Arduino board.

In this project, the following header files are required in the main_code.ino for programming:

`#include <Wire.h>`. This header file is especially required for I2C protocol devices for faster communication. In this project, the two I2C devices are SSD1306 OLED display and DS3231 RTC module. (This header file is already included in the latest Arduino IDE.)

`#include <Adafruit_GFX.h>`. The header file Adafruit Graphical library is used to generate different types of graphics like square, triangle, black and white colour inversions, star symbols, and images in .bmp formats. In this project, different font sizes and special symbols notation for temperature units (degrees

centigrade) are used.

`#include <Adafruit_SSD1306.h>`. This header file is used to communicate with the SSD1306 display using address mapping. The SSD1306 display having 128 × 64 pixel size can display the data up to eight pages alternately. Access each page by using three address lines. The selection of address lines, data and commands are very easy using Adafruit_SSD1306 header file.

The Arduino code (main_code.ino) requires two external libraries: Adafruit-GFX-Library-master.zip and Adafruit_SSD1306-master.zip. You need to include them in Arduino IDE before compiling and uploading the code to the Arduino board.

An actual-size PCB layout for Arduino real-time clock with temperature display is shown in Fig. 7 and its components layout in Fig. 8.

After assembling the circuit on the PCB, connect Arduino Uno pins to PCB using jumper wires. Do not forget to upload the source code into the Arduino Uno board before wiring. The circuit works with 230V AC mains using the transformer (X1), as shown in the circuit. **EFY**



Pamarthi Kanakaraja is an assistant professor (R&D cell) at KL University, Vaddeswaram, Guntur district, Andhra Pradesh



Dr K Sarat Kumar is a professor at KL University. He is also one of the technical members from KL University, providing technical support on RF antenna designs to NARL (National Atmospheric Research Laboratory). He has 22 years of work experience in RF antenna designing, testing and simulation in a real-time environment

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